Uch o'lchamli panjarada fazoviy jismlarni chizish

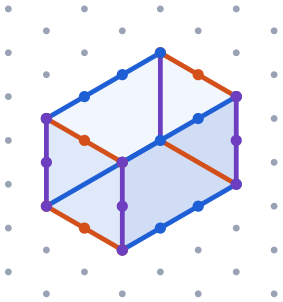
2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

---

1. **Rasmda kuboidning hamma qirralari chizilgan. Ortiqcha chiziqlarning umumiy uzunligi necha birlik?**

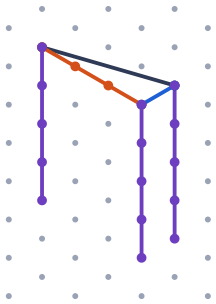


A) 3 **B)** 28

C) 21 **D)** 7

---

2. **Prizma chizmasi 2-qadamgacha chizilgan (rasmda). Chizma tugagach, unda jami nechta chiziq bo'ladi?**

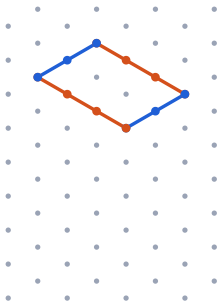


A) 8 **B)** 7

C) 9 **D)** 6

---

3. **Kuboidning ustki yog'i chizilgan (rasmda). Tugatish uchun yana nechta chiziq kerak?**

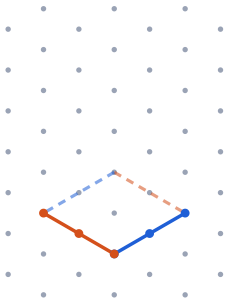


A) 4 **B)** 5

C) 7 **D)** 8

---

4. **Kvadrat asosli piramidaning asosi chizilgan (orqa qirralari punktir). Chizmani tugatish uchun uchidan nechta chiziq tortiladi?**



A) 3 **B)** 2

C) 5 **D)** 4

---

5. **Qirrasi 2 birlik bo'lgan kub chizildi. Chizmadagi chiziqlarning umumiy uzunligi necha birlik?**

A) 24 **B)** 18 **C)** 16 **D)** 9

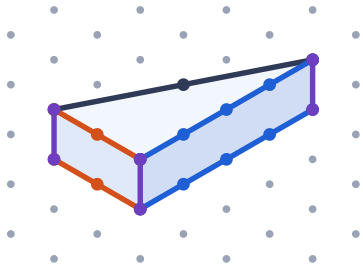
---

6. **Bo'yi 5, eni 1, balandligi 2 birlik bo'lgan kuboid chizildi. Chizmadagi chiziqlarning umumiy uzunligi necha birlik?**

A) 32 **B)** 24 **C)** 8 **D)** 16

---

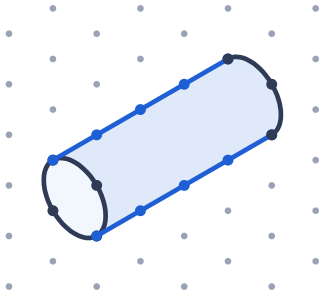
7. **Rasmdagi prizmaning asosidagi ikki qirrasi va balandligi necha birlik?**



A) 4 va 4, balandligi 1
B) 3 va 5, balandligi 2
C) 2 va 1, balandligi 4
D) 2 va 4, balandligi 1

---

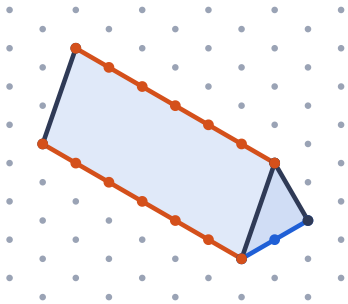
8. **Rasmdagi yotgan silindrning uzunligi necha birlik?**



A) 4 **B)** 5
C) 2 **D)** 3

---

9. **Rasmdagi prizma chizmasida chapga-qiya chiziqlar nechta va har biri necha birlik?**



A) 3 ta, 7 birlikdan **B)** 3 ta, 6 birlikdan
C) 2 ta, 7 birlikdan **D)** 2 ta, 6 birlikdan

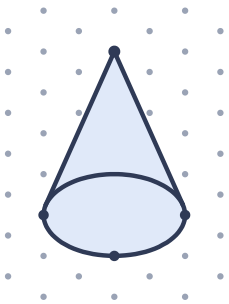
---

10. **Silindr va konusning asosi bir xil oval, balandligi ham bir xil. Konus chizmasida silindrnikiga qaraganda nima yo'q?**

A) Pastki ovalning oldingi yarmi **B)** Ikkita chiziq
C) Panjara nuqtalari **D)** Ustki oval

---

11. **Aziz konusni chizdi (rasmda). Uning chizmasida nima ortiqcha?**



A) Uchidan tortilgan chiziqlar
B) Konusning uchi
C) Asos ovalining orqa yarmi
D) Asos ovalining oldingi yarmi

---

12. **Qaysi jism chizmasida oval ham, to'g'ri chiziq ham bor, lekin parallel chiziqlar yo'q?**

A) Prizma **B)** Piramida **C)** Silindr **D)** Konus

---

13. **Bo'yi 4, eni 2, balandligi 3 birlik bo'lgan kuboid chizildi. Eni bo'yicha (o'ngga-qiya) chiziqlar nechta va har biri necha birlik?**

A) 4 ta, 2 birlikdan **B)** 3 ta, 2 birlikdan
C) 2 ta, 2 birlikdan **D)** 3 ta, 4 birlikdan

---

14. **Tik turgan prizmaning asosi uchburchak, balandligi 5 birlik. Chizmada uchala tik qirra ko'rinadi. Tik chiziqlarning umumiy uzunligi necha birlik?**

A) 5 **B)** 20 **C)** 10 **D)** 15

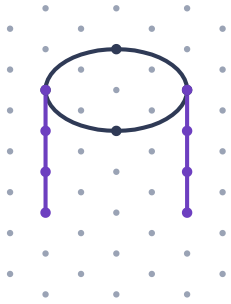
---

15. **Konusning balandligi 4 birlik. Oval o'rtasidan uchigacha bo'lgan tik chiziq uchlari bilan nechta panjara nuqtasidan o'tadi?**

A) 4 **B)** 6 **C)** 3 **D)** 5

---

16. **Silindr chizmasi rasmda. Uni tugatish uchun nima chizish kerak?**



A) Ustki oval

B) Yana ikkita tik chiziq

C) Butun pastki oval

D) Pastki ovalning oldingi yarmi

---

17. **Yotgan silindrning old ovali chizilgan, unda 4 ta panjara nuqtasi belgilangan (rasmda). Parallel chiziqlar qaysi nuqtalardan tortiladi?**



A) Belgilangan 4 ta nuqtaning hammasidan

B) Eng chap va eng o'ng belgilangan nuqtadan

C) Eng yuqori va eng past belgilangan nuqtadan

D) Ovalning o'rtasidan

---

18. **Kuboid chizmasidagi chiziqlarning umumiy uzunligi 30 birlik. Kuboidning bo'yi 4, eni 3 birlik. Balandligi necha birlik?**

A) 3 **B)** 10 **C)** 7 **D)** 23

---

19. **Tom shaklidagi prizmaning uzunligi 3 birlik edi. Uni 5 birlik qilib chizsak, chizmada nima o'zgaradi?**

A) Faqat parallel chiziqlar uzayadi

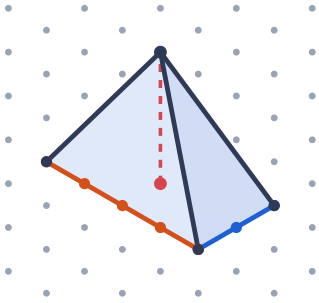
B) Uchburchak kattalashadi

C) Hamma chiziqlar uzayadi

D) Hech narsa o'zgarmaydi

---

20. **Rasmdagi piramidaning balandligi (qizil punktir) va asosi necha birlik?**



A) Asosi 2 × 4, balandligi 2

B) Asosi 4 × 2, balandligi 3

C) Asosi 4 × 2, balandligi 4

D) Asosi 5 × 3, balandligi 4

Javoblar

Uch o'lchamli panjarada fazoviy jismlarni chizish · 2-test (qiyinroq)

№	Javob	Qiymati
1	**D**	7
2	**A**	8
3	**B**	5
4	**A**	3
5	**B**	18
6	**B**	24
7	**D**	2 va 4, balandligi 1
8	**A**	4
9	**D**	2 ta, 6 birlikdan
10	**D**	Ustki oval

№	Javob	Qiymati
11	**C**	Asos ovalining orqa yarmi
12	**D**	Konus
13	**B**	3 ta, 2 birlikdan
14	**D**	15
15	**D**	5
16	**D**	Pastki ovalning oldingi yarmi
17	**C**	Eng yuqori va eng past belgilangan nuqtadan
18	**A**	3
19	**A**	Faqat parallel chiziqlar uzayadi
20	**B**	Asosi 4 × 2, balandligi 3

Qisqa yechimlar

1. Orqadagi 3 ta qirra ortiqcha — har yo'nalishdan bittadan.
Oraliqlarni sanaymiz: o'ngga-qiya 3, chapga-qiya 2, tik 2 birlik.
$3 + 2 + 2 = 7$. Javob: **7** birlik.

---

2. Hozir: ustki uchburchak — 3 ta, tik chiziqlar — 3 ta, jami 6 ta.
Pastki uchburchakdan faqat 2 ta qirra ko'rinadi, uchinchisi orqada.
$6 + 2 = 8$. Javob: **8** ta.

---

3. Tik qirralar: 4 tadan 3 tasi ko'rinadi. Pastki yoq: 4 ta qirradan 2 tasi ko'rinadi.
$3 + 2 = 5$. Kerak: **5** ta chiziq.

---

4. Asosning 4 ta uchidan biri orqada: unga tortilgan chiziq ko'rinmaydi.
$4 - 1 = 3$. Javob: **3** ta.

---

5. Chizmada 9 ta qirra bor, har biri 2 birlik.
$9 \cdot 2 = 18$. Javob: **18** birlik.
(24 — 12 ta qirraning hammasi olinganda chiqadi.)

---

6. Chizmada har yo'nalishda 3 tadan qirra bor.
$3 \cdot 5 + 3 \cdot 1 + 3 \cdot 2 = 15 + 3 + 6 = 24$.
Javob: **24** birlik.

---

7. Ustki uchburchak: chapga-qiya 2 oraliq, o'ngga-qiya 4 oraliq. Tik qirralar — 1 oraliq.
Javob: **2 va 4, balandligi 1**.
(3 va 5 — nuqtalar sanalganda chiqadi.)

---

8. Parallel chiziqdagi oraliqlarni sanaymiz: 4 ta (nuqtalar 5 ta).
Uzunligi **4** birlik.

---

9. Uchburchakning uch uchidan biri orqada: undan chiqqan chiziq ko'rinmaydi. Ko'rinadigani — 2 ta.
Har birida 6 ta oraliq.
Javob: **2 ta, 6 birlikdan**.

---

10. Silindrda: ustki oval, pastki ovalning oldingi yarmi va 2 ta tik chiziq.
Konusda: pastki ovalning oldingi yarmi va uchiga boruvchi 2 ta chiziq.
Konusda **ustki oval** yo'q.

---

11. Konusning asosi pastda, uchi tepada. Asos ovalining orqa yarmi konus orqasida qoladi — u ko'rinmaydi.
Ortiqcha: **asos ovalining orqa yarmi**.

---

12. Oval faqat silindr va konusda bor. Silindrning ikki chizig'i parallel.
Konusning ikki chizig'i bitta uchda uchrashadi.
Javob: **konus**.

---

13. Kuboidning eni bo'yicha 4 ta qirrasidan biri orqada qoladi: ko'rinadigani 3 ta.
Har biri eniga teng — 2 birlik.
Javob: **3 ta, 2 birlikdan**.

---

14. 3 ta tik chiziq, har biri 5 birlik.
$3 \cdot 5 = 15$. Javob: **15** birlik.

---

15. 4 birlik — 4 ta oraliq. Nuqtalar bittaga ko'p.
$4 + 1 = 5$. Javob: **5** ta nuqta.

---

16. Ustki oval va 2 ta tik chiziq chizilgan. Qolgani — pastki asos.
Uning orqa yarmi ko'rinmaydi.
Kerak: **pastki ovalning oldingi yarmi**.

---

17. Belgilangan nuqtalar — tik turgan rombning uchlari. Silindr chapga-qiya yotibdi, ovalga tegib o'tadigan ikki parallel chiziq rombning eng yuqori va eng past uchidan chiqadi.
Javob: **eng yuqori va eng past belgilangan nuqtadan**.

---

18. Chizmada har yo'nalishda 3 tadan qirra bor.
Bo'yi va eni bo'yicha: $3 \cdot 4 + 3 \cdot 3 = 21$ birlik.
3 ta tik qirraga: $30 - 21 = 9$ birlik. Balandligi $9 : 3 = 3$.
Javob: **3** birlik.

---

19. Uchburchak — asos, u o'zgarmaydi. Uzunlik — asoslarni tutashtiruvchi parallel chiziqlar.
Javob: **faqat parallel chiziqlar uzayadi**.

---

20. Asos: chapga-qiya 4 oraliq, o'ngga-qiya 2 oraliq. Qizil punktirda 3 ta oraliq.
Javob: **asosi 4 × 2, balandligi 3**.
